Duals of Hilbert Spaces; Riesz Representation Theorem; Operators in Hilbert Spaces

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Contents

Objectives

- Study the adjoint of an operator
- Study (Self-adjoint, normal, unitary operators)
- Study the spectrum of an operator.

0.1 Riesz Representation Theorem

Theorem 1 (Riesz Representation Theorem)**.** *Let H be a Hilbert space. For any bounded linear functional $\varphi \in H^*$, there exists a unique element $y \in H$ such that*

$$\varphi(x) = \langle x, y\rangle$$

for all $x \in H$ and $\|\varphi\| = \|y\|$

Now recall that $\mathbb{R}^n$, (ℓ_2 and $L_2[a, b]$ are Hilbert spaces. Using the Riesz representation theorem, we can know the nature of bounded linear functionals on these spaces as in the following examples.

Example 1. *For $F \in (\mathbb{R}^n)^*$, there exists a unique $a = (a_1, a_2, a_3, \dots, a_n) \in \mathbb{R}^n$ such that*

$$F(x) = \langle x, a\rangle = \sum_{i=1}^{n} x_i a_i, \text{ for all } x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

Example 2. *For $F \in (\ell_2)^*$, there exists a unique $a = (a_1, a_2, a_3, \dots, a_n, \dots) \in \ell_2$ such that*

$$F(x) = \langle x, a\rangle = \sum_{i=1}^{\infty} x_i \overline{a_i}, \text{ for all } x = (x_1, x_2, \dots, x_n, \dots) \in \ell_2$$

Example 3. *For $F \in (L_2[a, b])^*$, there exists a unique $g \in L_2[a, b]$ such that*

$$F(f) = \langle f, g\rangle = \int_a^b f(t)\overline{g}(t)dt, \text{ for all } f \in L_2$$

Corollary 1. *Every Hilbert space is reflexive.*

Corollary 2. *Let H be a Hilbert space and $T \in B(H)$, then*

$$\|T\| = \sup\{|\langle T(x), y\rangle| : x, y \in X \text{ and } \|x\| = \|y\| = 1\}.$$

Proof. Let $\alpha = \sup\{|\langle T(x), y\rangle| : x, y \in H$ and $\|x\| = \|y\| = 1\}$. By the Cauchy-Schwarz inequality, we have $|\langle T(x), y\rangle| \leq \|T(x)\|\|y\|$ and by the definition of the operator norm we get $\|T(x)\|\|y\| \leq \|T\|\|x\|\|y\|$. Combining these two inequalities and substituting $\|x\| = \|y\| = 1$, we obtain $|\langle T(x), y\rangle| \leq \|T\|$. Taking the supremum, we reach $\alpha \leq \|T\|$.

Now, let $x \in H$ be such that $\|x\| = 1$ and $T(x) \neq 0$. (If $T(x) = 0$ $\forall \|x\| = 1$, then $\|T\| = 0$ and so $\|T\| \leq \alpha$). As $T(x) \neq 0$, by one of the consequences of the Hahn-Banach Theorem (Corollary 6.1 from the notes), there exists $f \in H^*$ continuous such that $\|f\| = 1$ and $f(T(x)) = \|T(x)\|$. Now since $f \in H^*$ and f is bounded, by the Riesz Representation Theorem, there exists $y_f \in H$ such that $\|y_f\| = \|f\| = 1$ and $\|T(x)\| = f(T(x)) = \langle T(x), y_f \rangle$. Therefore, $\|T(x)\| \leq \alpha$ for all $\|x\| = 1$. This implies $\|T\| \leq \alpha$.

Thus, $\|T\| = \sup\{|\langle T(x), y\rangle| : x, y \in X \text{ and } \|x\| = \|y\| = 1\}$ □

Corollary 3 (The weak compactness property)**.** *Every bounded sequence in a Hilbert space H containing a weakly convergent subsequence.*

0.2 The Adjoint

Definition 1 (Adjoint Operator)**.** *Let H and G be Hilbert spaces and let $T : H \longrightarrow G$ be a bounded linear operator. An operator $T^* : G \to H$ such that*

$$\langle T(x), y \rangle_G = \langle x, T^*(y) \rangle_H \tag{1}$$

for all $x \in H, y \in G$, is called the adjoint of T.

Theorem 2. *For any bounded linear operator T on a Hilbert space H, the adjoint operator T^* exists and is unique.*

Proof. **Existence of the Adjoint:** We would like to show that for all $y \in G$, there is $w \in H$ such that

$$\langle Tx, y \rangle = \langle x, w \rangle$$

for all $x \in H$. Let

$$\begin{aligned} f_y : H &\longrightarrow \mathbb{F} \\ x &\longmapsto \langle Tx, y \rangle \end{aligned}$$

This map is linear, since it is the composition of two linear maps, namely T and $\langle \cdot, y \rangle$. Also, this map is bounded since

$$\begin{aligned} |f_y(x)| = |\langle Tx, y \rangle| &\leq \|y\| \|Tx\| \\ &\leq \|y\| \|T\| \|x\| \end{aligned}$$

So, $f_y \in H^*$. By Reisz Representation Theorem, for some $w \in H$, we can write f_y as

$$\langle Tx, y \rangle = f_y(x) = \langle x, w \rangle$$

Uniqueness of the Adjoint: Let $S : H \to G$ and $S' : H \to G$ be adjoints of T, i.e. for all $x, y \in G$

$$\begin{aligned} \langle Tx, y \rangle &= \langle x, Sy \rangle \\ \langle Tx, y \rangle &= \langle x, S'y \rangle \end{aligned}$$

We then have

$$\langle x, (S - S')y \rangle = \langle x, Sy \rangle - \langle x, S'y \rangle = \langle Tx, y \rangle - \langle Tx, y \rangle = 0$$

This implies that $(S - S')y = 0$, and hence, $S - S' = 0 \implies S = S'$. □

The defining relation of the adjoint (1) shows that the action of A on H_0 is mimicked by the action of the adjoint A^* on H_1; this mimicry is visible only through the applicable inner products, so the adjoint itself depends on these inner products. Roughly, if A has some effect, A^* preserves the geometry of that effect while acting with a reversed domain and codomain.

Example 4. *Let I be the identity operator on a Hilbert space H. then $I^* = I$*

Example 5. *For $k \in C[0,1]$, let $T_k : C[0,1] \to C[0,1]$ be given by $(T_k g)(t) = k(t)g(t)$. Then $(T_k)^* = T_{\bar{k}}$.*

Example 6. *Let $T_1 : H \to H$ be given by $T_1(x) = \alpha x$ for some scalar α. For any x and y in H,*

$$\langle T_1(x), y\rangle = \langle \alpha x, y\rangle = \alpha\langle x, y\rangle = \langle x, \bar{\alpha} y\rangle$$

where the first equality follows from the definition of T_1, the second from the linearity in the first component of the inner product, and the third from the conjugate linearity in the second component of the inner product. Comparing with (1), the adjoint of T_1 is $T_1^*(y) = \bar{\alpha} y$. Hence, the adjoint of multiplication by a scalar is multiplication by the conjugate of the scalar.

From the above example it follows that if H is a real Hilbert space, then $T_1 = T_1^*$.

Example 7. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear operator represented by the matrix $A \in \mathbb{R}^{m\times n}$, i.e. $T(x) = Ax$. The matrix of the adjoint of T is the transpose matrix $A^T \in \mathbb{R}^{n\times m}$, since for any $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$:*

$$\langle x, T(y)\rangle_{\mathbb{R}^m} = \langle x, Ay\rangle_{\mathbb{R}^m} = x \cdot (Ay) = x^t(Ay) = (x^t A)y = (A^t x)^t y = (A^t x) \cdot y.$$

In component notation, this identity is expressed by swapping the order of summation:

$$\sum_{i=1}^{m} x_i \left(\sum_{j=1}^{n} a_{ij} y_j\right) = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} x_i\right) y_j.$$

For a linear map on complex spaces $T : \mathbb{C}^n \to \mathbb{C}^m$, the matrix of the adjoint with respect to the standard inner product is the Hermitian conjugate matrix, $A^* = \overline{A^T}$, satisfying $\langle x, Ay\rangle = \langle A^* x, y\rangle$.

Example 8. *Suppose that S and T are the right and left shift operators, respectively, on the sequence space $l^2(\mathbb{R})$, defined by*

$$S(x_1, x_2, x_3, \dots) = (0, x_1, x_2, x_3, \dots), \quad T(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots).$$

Then $S^* = T$, since

$$\langle T(x), y\rangle = x_1(0) + x_2 y_1 + x_3 y_2 + x_4 y_3 + \cdots = \langle x, S(y)\rangle.$$

Also, $T^* = S$, since

$$\langle S(x), y\rangle = (0)y_1 + x_1 y_2 + x_2 y_3 + x_3 y_4 + \cdots = \langle x, T(y)\rangle.$$

Theorem 3. *Let H, G be Hilbert spaces and let $T : H \to G$ and $S : H \to G$ be linear operator, then*

1. $(T^*)^* = T$

2. $(T+S)^* = T^* + S^*$

3. $(\alpha T)^* = \bar{\alpha} T^*$

4. $(TS)^* = S^* T^*$

Proof. 1. We have

$$\langle (T^*)^* x, y\rangle = \overline{\langle y, (T^*)^* x\rangle} = \overline{\langle T^* y, x\rangle} = \langle x, T^* y\rangle = \langle Tx, y\rangle$$

Therefore, $(T^*)^* = T$.

2. We have

$$\begin{aligned}\langle x, (T+S)^* y\rangle = \langle (T+S)x, y\rangle &= \langle Tx, y\rangle + \langle Sx, y\rangle \\ &= \langle x, T^* y\rangle + \langle x, S^* y\rangle \\ &= \langle x, (T^* + S^*) y\rangle\end{aligned}$$

Therefore, $(T+S)^* = T^* + S^*$.

3. We have

$$\begin{aligned}\langle x, (\alpha T)^* y\rangle = \langle \alpha T x, y\rangle = \alpha\langle Tx, y\rangle = \alpha\langle x, T^* y\rangle &= \alpha\overline{\langle T^* y, x\rangle} \\ &= \overline{\bar{\alpha}\langle T^* y, x\rangle} \\ &= \overline{\langle \bar{\alpha} T^* y, x\rangle} \\ &= \langle x, \bar{\alpha} T^* y\rangle\end{aligned}$$

Therefore, $(\alpha T)^* = \bar{\alpha} T^*$.

4. We have

$$\langle x, (TS)^* y\rangle = \langle TSx, y\rangle = \langle Sx, T^* y\rangle = \langle x, S^* T^* y\rangle$$

Therefore, $(TS)^* = S^* T^*$.

□

Proposition 1. *If H is a complex Hilbert space and $T \in B(H)$ is invertible then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.*

Proof. As $TT^{-1} = T^{-1}T = I$, if we take the adjoint of this equation we obtain $(TT^{-1})^* = (T^{-1}T)^* = I^*$ and so $(T^{-1})^* T^* = T^* (T^{-1})^* = I$ (see example below). Hence T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$. □

Theorem 4. *If $T \in B(H, G)$, then T^* is a bounded linear operator with $\|T^*\| = \|T\|$.*

Proof. First, T^* is linear, since, for all $x \in H$

$$\begin{aligned}\langle x, T^*(\alpha u + \beta v)\rangle = \langle T(x), \alpha u + \beta v\rangle &= \overline{\langle \alpha u + \beta v, T(x)\rangle} \\ &= \overline{\alpha\langle u, T(x)\rangle + \beta\langle v, T(x)\rangle} \\ &= \overline{\alpha\langle T^*(u), x\rangle + \beta\langle T^*(v), x\rangle} \\ &= \overline{\langle \alpha T^*(u) + \beta T^*(v), x\rangle} \\ &= \langle x, \alpha T^*(u) + \beta T^*(v)\rangle\end{aligned}$$

Now, we show that T^* is bounded. Consider the following map

$$\begin{aligned} \varphi : H &\longrightarrow \mathbb{F} \\ x &\longmapsto \langle x, T^* y\rangle \end{aligned}$$

We see that

$$|\varphi(x)| = |\langle x, T^* y\rangle| = |\langle Tx, y\rangle| \leq \|y\|\|Tx\| \leq \|y\|\|T\|\|x\|$$

So, φ is bounded and $\|\varphi\| \leq \|T\|\|y\|$. By Reisz Representation Theorem, we have $\|T^*y\| = \|\varphi\| \leq \|T\|\|y\|$. Therefore, T^* is bounded with $\|T^*\| \leq \|T\|$. Now, consider the map

$$\begin{aligned} \tilde{\varphi} : G &\longrightarrow \mathbb{F} \\ y &\longmapsto \langle y, Tx\rangle \end{aligned}$$

We have

$$|\tilde{\varphi}(y)| = |\langle y, Tx\rangle| = |\overline{\langle Tx, y\rangle}| = |\langle Tx, y\rangle| = |\langle x, T^* y\rangle| \leq \|x\|\|T^* y\| \leq \|x\|\|T^*\|\|y\|$$

So, $\tilde{\varphi}$ is bounded and $\|\tilde{\varphi}\| \leq \|T^*\|\|x\|$. By Reisz Representation Theorem, $\|Tx\| = \|\tilde{\varphi}\| \leq \|T^*\|\|x\|$. Therefore, $\|T\| \leq \|T^*\|$, and hence, $\|T^*\| = \|T\|$. □

Proposition 2. *Let H, G be Hilbert spaces, and $T : H \to G$. then*

1. $\|T^*T\| = \|T\|^2$
2. $\ker T = (\operatorname{Im} T^*)^\perp$
3. $\ker T^* = (\operatorname{Im} T)^\perp$
4. $\ker T^* = \{0\} \Leftrightarrow \operatorname{Im} T$ *dense in G.*

Proof. 1. Firstly, we have

$$\|T^*Tx\| \leq \|T^*\|\|Tx\| \leq \|T^*\|\|T\|\|x\|$$

So, $\|T^*T\| \leq \|T^*\|\|T\| = \|T\|\|T\| = \|T\|^2$. We also observe that

$$\begin{aligned} \|Tx\|^2 = \langle Tx, Tx\rangle = \langle x, T^*Tx\rangle \leq \|x\|\|T^*Tx\| &\leq \|x\|\|T^*T\|\|x\| \\ &= \|T^*T\|\|x\|^2 \end{aligned}$$

So, $\|T\|^2 \leq \|T^*T\|$, and therefore, $\|T^*T\| = \|T\|^2$

2. Let $u \in \ker T$, i.e. $Tu = 0$. This is true if and only if $\langle Tu, v\rangle = 0$ for all $v \in G$. So, $\langle u, T^*v\rangle = \langle Tu, v\rangle = 0$. $\langle u, T^*v\rangle = 0$ if and only if u is orthogonal to every element of $\operatorname{Im} T$, i.e. $u \in (\operatorname{Im} T^*)^\perp$.

3. By the second property above, we have

$$\ker T^* = (\operatorname{Im}(T^*)^*)^\perp = (\operatorname{Im} T)^\perp$$

4. If $\ker T^* = \{0\}$, then, by the third property above, $(\operatorname{Im} T)^\perp = \{0\}$. Here, $(\operatorname{Im} T)^\perp = \{0\}$ if and only if $\operatorname{Im} T$ is dense in G.

□

Theorem 5. *Let H be a Hilbert space and $T : H \to H$ then T is invertible $\iff \ker T^* = \{0\}$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha\|x\| \quad \forall x \in H$.*

Proof. $\implies$) Assume that T is invertible, then T^* is invertible and $T^{-1} \in B(\mathcal{H})$. So, $\ker T^* = \{0\}$ and $\|T^{-1}u\| \leq \|T^{-1}\|\|u\| = \|T^{-1}\|\|TT^{-1}u\| \implies \|TT^{-1}u\| \geq \frac{1}{\|T^{-1}\|}\|T^{-1}u\|$. Letting $x = T^{-1}u$, we have $\ker T^* = \{0\}$ and $\|Tx\| \geq \frac{1}{\|T^{-1}\|}\|x\|$.
$\impliedby$) Assume that $\ker T^* = \{0\}$ and $\|Tx\| \geq \alpha\|x\|$ for some $\alpha \in \mathbb{F}$. T is injective, since if $Tx = 0$, then

$$0 = \|Tx\| \geq \alpha\|x\| \implies \|x\| \leq 0 \implies x = 0$$

Now, we show that T is surjective. Since $\ker T^* = \{0\}$, by 2, $\operatorname{Im} T$ is dense in $\mathcal{H}$. So, for every $y \in \mathcal{H}$, there is a sequence $\{Tx_n\} \in \operatorname{Im} T$ such that $Tx_n \to y$. So, $\{Tx_n\}$ is a Cauchy sequence. So, we have

$$\|Tx_n - Tx_m\| \geq \alpha\|x_n - x_m\| \implies \|x_n - x_m\| \leq \frac{1}{\alpha}\|Tx_n - Tx_m\|$$

So, $\{x_n\} \in \mathcal{H}$ is a Cauchy sequence. Since $\mathcal{H}$ is a Hilbert space, $\{x_n\}$ converges to some $x \in \mathcal{H}$. So, $Tx_n \to Tx$. By uniqueness of limit, and that $Tx_n \to y$, $Tx = y$. So, T is surjective and T is invertible. □

Corollary 4. *Let H be a Hilbert space and $T : H \to H$ then T is invertible $\iff \exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha\|x\| \quad \forall x \in H$.*

Proof. If T is invertible, then so is T^*. So, by Theorem 5, $\exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha\|x\|$ for all $x \in H$. Conversely, if $\exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$, then if $T^*u = 0$ for some $u \in H$, then

$$\begin{aligned} 0 = \|T^*u\| &\geq \beta\|u\| \\ &\implies u = 0 \end{aligned}$$

So, $\ker T^* = \{0\}$. By Theorem 5, T is invertible. □

Example 9. *Let $T : l^2 \to l^2$ where*

$$T(x_1, x_2 \ldots) = (0, x_1, x_2 \ldots)$$

As shown in Example 8, we have $T^*(x_1, x_2, \ldots) = (x_2, x_3, \ldots)$. So,

$$\ker T^* = \{u \in \mathbb{F} : (u, 0, 0, \ldots)\} \neq \{(0, 0, 0, \ldots)\}$$

By Theorem 5, T is not invertible.

Proposition 3. *Let H be a complex Hilbert space and let $T \in B(H)$, then Ker T = Ker (T^*T). Moreover, the closure of Im T^* is equal to the closure of Im (T^*T), where Im T^* denotes the image of T^*.*

Theorem 6. *Suppose that $A : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator on a Hilbert space $\mathcal{H}$ with closed range. Then the equation $Ax = y$ has a solution for x if and only if y is orthogonal to $\ker A^*$.*

Example 10. *Consider the multiplication operator $M : L^2([0, 1]) \to L^2([0, 1])$ defined by*

$$Mf(x) = xf(x).$$

Then $M^* = M$, and M is one-to-one, so every $g \in L^2([0, 1])$ is orthogonal to $\ker M^*$; but the range of M is a proper dense subspace of $L^2([0, 1])$, so $Mf = g$ is not solvable for every $g \in L^2([0, 1])$.

0.3 Normal Operators

Definition 2. *Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \to \mathcal{H}$ be a linear operator. Then we said T is* ***normal*** *if $TT^* = T^*T$*

Example 11. *Consider*

$$\begin{aligned} T : \mathbb{R}^2 &\longrightarrow \mathbb{R}^2 \\ x &\longmapsto \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} x \end{aligned}$$

We see that

$$T^*(x) = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^T x = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} x$$

Since

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

we have $T^*T = TT^*$.

Example 12. *If T is the right shift operator defined in Example 8, then T is not normal. This can be shown by finding T^*T and TT^*. We have*

$$T^*T(x_1, x_2, \dots) = T^*(0, x_1, x_2, \dots) = (x_1, x_2, \dots)$$

and

$$TT^*(x_1, x_2, \dots) = T(x_2, x_3, \dots) = (0, x_2, x_3, \dots)$$

which are not equal.

Example 13. *$T_k : L^2[0, 1] \to L^2[0, 1]$, defined by*

$$(T_k f)(t) = k(t)f(t), \qquad k \in C[0, 1],$$

is a ***normal*** *operator. Since $T_k^* = T_{\bar{k}}$, we have*

$$T_k T_k^* f = T_k(T_{\bar{k}} f) = k\bar{k}f$$

and

$$T_k^* T_k f = T_{\bar{k}}(T_k f) = \bar{k}kf$$

for all $f \in L^2[0, 1]$. Hence $T_k T_k^* = T_k^* T_k$.

Example 14. *If T is normal, then T - λI is also normal. That is, since*

$$\begin{aligned} (T - \lambda I)^*(T - \lambda I) = (T^* - \bar{\lambda} I)(T - \lambda I) &= T^*T - \bar{\lambda}T - \lambda T^* + \bar{\lambda}\lambda I \\ &= TT^* - \bar{\lambda}T - \lambda T^* + \lambda\bar{\lambda} I \\ &= T(T^* - \bar{\lambda} I) - \lambda(T^* - \bar{\lambda} I) \\ &= (T - \lambda I)(T^* - \bar{\lambda} I) \\ &= (T - \lambda I)(T - \lambda I)^* \end{aligned}$$

Lemma 1. *Let $T : H \to H$ be a normal operator, then*

1. $\|T(x)\| = \|T^*(x)\|$; $\quad \forall x \in H$

2. $\ker(T) = \ker(T^*)$

3. *If there is* $\alpha > 0$ *such that* $\|T(x)\| \geqslant \alpha\|x\| \quad \forall x \in H$ *then* $\ker T^* = \{0\}$.

Proof. 1. We have

$$\|Tx\| = \langle Tx, Tx\rangle = \langle x, T^*Tx\rangle = \langle x, TT^*x\rangle = \langle T^*x, T^*x\rangle = \|T^*x\|$$

2. Let $x \in \ker(T)$. Then $Tx = 0$, so $\|Tx\| = 0$. By 1, since T is normal, we have $\|Tx\| = \|T^*x\|$. Hence $\|T^*x\| = 0$, which implies $T^*x = 0$. Thus $x \in \ker(T^*)$, and so

$$\ker(T) \subseteq \ker(T^*).$$

In a similar way, one shows that

$$\ker(T^*) \subseteq \ker(T).$$

Therefore, $\ker(T) = \ker(T^*)$.

3. Let $u \in \ker T^*$, i.e. $T^*(u) = 0$. By 1, we have

$$0 = \|T^*u\| = \|Tu\| \geq \alpha\|u\| \geq 0$$

Hence, $\|u\| = 0 \implies u = 0$

□

By the lemma above and theorem 5, the next theorem immediately follows.

Theorem 7. *Let* H *be a Hilbert space and* $T : H \to H$ *is a normal operator, then* T *is invertible* $\iff \exists\alpha >$ *such that* $\|T(x)\| \geqslant \alpha\|x\| \forall x \in H$.

Hence it is easier to determine whether a normal operator invertible compared with any operator.

Theorem 8. *Let* T *be a bounded* ***normal*** *operator on a Hilbert space* $\mathcal{H}$. *Then* T *is* ***invertible*** *if and only if* $\ker(T) = \{0\}$ *and the range of* T *is closed. Moreover, if* T *is invertible, then* T^{-1} *is bounded and normal.*

Proof. $(\Rightarrow)$ Assume T is invertible. Then T is one-to-one and onto. Since T is one-to-one, we have $\ker(T) = \{0\}$. Since T is onto, we have $\text{Range}(T) = \mathcal{H}$, which is closed.

$(\Leftarrow)$ Assume $\ker(T) = \{0\}$ and $\text{Range}(T)$ is closed. Now we show that T is onto. $\overline{\text{Range}(T)} = (\ker(T^*))^\perp$ and $\text{Range}(T)$ is closed, it follows that

$$\text{Range}(T) = (\ker(T^*))^\perp.$$

Since T is normal, $\ker(T^*) = \ker(T) = \{0\}$. Hence $\text{Range}(T) = (\{0\})^\perp = \mathcal{H}$. Therefore, T is onto. Thus T is bijective. Since $\mathcal{H}$ is a Banach space, the bounded inverse theorem implies that T^{-1} is bounded.

Now suppose T is invertible. Therefore,$T^{-1}(T^{-1})^* = T^{-1}(T^*)^{-1} = (T^*T)^{-1} = (TT^*)^{-1} = (T^{-1})^*T^{-1}$ so T^{-1} is normal. □

Corollary 5. *Let T be a bounded **normal** operator on a Hilbert space $\mathcal{H}$. Then the following statements are equivalent:*

1. *T is invertible*
2. *there exists $c > 0$ such that $\|T(x)\| \geq c\|x\|$ for all $x \in \mathcal{H}$*
3. *$\ker(T) = \{0\}$ and the range of T is closed.*

Example 15. *Consider the diagonal operator $D : \ell^2 \to \ell^2$ defined by*

$$D(x_1, x_2, x_3, \dots) = \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots\right) = \left\{\frac{x_n}{n}\right\}_{n \geq 1}.$$

*Then D is **not** invertible and its range is **not** closed. **Method 1** $\neg(2) \implies \neg(1)$**: Assume for contradiction** that there exists $c > 0$ such that*

$$\|D(x)\| \geq c\|x\| \qquad \textit{for all } x \in \ell^2.$$

In particular, take $e_n = (0, 0, \dots, \underbrace{1}_{n}, 0, \dots)$. Then $D(e_n) = \left(0, 0, \dots, \underbrace{\frac{1}{n}}_{n}, 0, \dots\right)$, so $\|D(e_n)\| = \frac{1}{n}, \quad \|e_n\| = 1$. Hence

$$\frac{1}{n} \geq c \qquad \textit{for all } n \in \mathbb{N},$$

which is a contradiction. Thus there is no such constant $c > 0$. Therefore,D is not invertible. Moreover,

$$\ker(D) = \left\{x \in \ell^2 : \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots\right) = 0\right\} = \{0\}.$$

*since D is normal, $\ker(D) = \{0\}$, and D is **not** invertible, it follows that the range of D is **not** closed. **Method 2** $\neg(3) \implies \neg(1)$**:** We have already shown that $\ker(D) = \{0\}$, so it remains to show that $\mathrm{Range}(D)$ **is not closed.** To do this, we choose a sequence (y_n) in the range and show that it converges to an element not in the range.*

Consider

$$y_n = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, 0, 0, \dots\right) = D(x_n),$$

where

$$x_n = (\underbrace{1, 1, 1, \dots, 1}_{\textit{first } n}, 0, 0, \dots).$$

Since $x_n \in \ell^2$, we have $y_n \in \mathrm{Range}(D)$ for all n.

Now consider $y = \left(1, \frac{1}{2}, \frac{1}{3}, \dots\right)$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$, we have $y \in \ell^2$. Also, $y_n \to y$ in ℓ^2, since

$$\|y_n - y\|^2 = \sum_{k=n+1}^{\infty} \frac{1}{k^2} \to 0.$$

However, $y \notin \mathrm{Range}(D)$, because if $y = D(x)$, then necessarily

$$x = (1, 1, 1, \dots),$$

but $(1, 1, 1, \dots) \notin \ell^2$.

Thus $\mathrm{Range}(D)$ is not closed. Hence D is not invertible.

0.4 Self-adjoint operators

An interesting subset of the normal operators is the self-adjoint operators.

Definition 3. *$T : H \to H$ is called a self-adjoint if $T = T^*$*

Example 16. *For any $T \in B(H)$, we observe that TT^* and T^*T are self-adjoint, since*

$$(T^*T)^* = T^*(T^*)^* = T^*T$$

and

$$(TT^*)^* = (T^*)^*T^* = TT^*$$

Theorem 9. *Let H be Hilbert space & T_1 and T_2 be self-adjont oterator, then*

1. *$\alpha T_1 + \beta T_2$ is a self-adjon, $\alpha\beta \in \mathbb{R}$*
2. *The set*

$$\mathfrak{S} = \{T \in B(H) : T \text{ is self-adjoint}\} \subset B(H)$$

is a closed subset of $B(H)$.

Proof. 1. We have

$$(\alpha T_1 + \beta T_2)^* = (\alpha T_1)^* + (\beta T_2)^* = \bar{\alpha}T_1^* + \bar{\beta}T_2^* = \alpha T_1^* + \beta T_2^* = \alpha T_1 + \beta T_2$$

Hence, $\alpha T_1 + \beta T_2$ is self-adjoint.

2. Let $\{T_n\}$ be a convergent sequence of elements in $\mathfrak{S}$ which converges to $T \in B(H)$, then

$$\lim_{n\to\infty} \|T_n - T\| = 0$$

We see that

$$\lim_{n\to\infty} \|T_n - T^*\| = \lim_{n\to\infty} \|T_n^* - T^*\| = \lim_{n\to\infty} \|(T_n - T)^*\| = \lim_{n\to\infty} \|T_n - T\| = 0$$

So,

$$\lim_{n\to\infty} \|T - T^*\| \le \lim_{n\to\infty} \|T_n - T\| + \lim_{n\to\infty} \|T_n - T^*\| = 0 + 0 = 0$$

Hence, $T^* = T$ and $T \in \mathfrak{S}$. Therefore, $\mathfrak{S} \subset B(H)$ is a closed set.

□

Proposition 4. *Any operator $T \in B(H)$, $T = R + iS$ for R and S self-adjoint*

Proof. Let R and S be the following

$$R = \frac{T + T^*}{2} \qquad S = \frac{T - T^*}{2i}$$

then we have

$$R + iS = \Big(\frac{T + T^*}{2}\Big) + i\Big(\frac{T - T^*}{2i}\Big) = \frac{T + T^*}{2} + \frac{T - T^*}{2} = \frac{2T}{2} = T$$

and

$$R^* = \Big(\frac{T+T^*}{2}\Big)^* = \frac{(T+T^*)^*}{2} = \frac{T^*+(T^*)^*}{2} = \frac{T^*+T}{2} = \frac{T+T^*}{2} = R$$

$$S^* = \Big(\frac{T-T^*}{2i}\Big)^* = -\frac{(T-T^*)^*}{2i} = -\frac{T^*-(T^*)^*}{2i} = -\frac{T^*-T}{2i} = \frac{T-T^*}{2i} = S$$

□

Theorem 10. *Let T be a bounded linear operator on a Hilbert space $\mathcal{H}$. Then the following statements are equivalent:*

1. *T is self-adjoint.*
2. *$\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in \mathcal{H}$.*
3. *$\langle Tx, x\rangle = \langle x, Tx\rangle$ for all $x \in \mathcal{H}$.*

Proof. ($1 \iff 2$) By definition of the adjoint, $\langle Tx, y\rangle = \langle x, T^*y\rangle$ Hence $T = T^*$ if and only if $\langle Tx, y\rangle = \langle x, Ty\rangle$

Also, ($2 \implies 3$) is clear by taking $y = x$.

Now we show ($3 \implies 2$). Assume that

First take $z = x + y$. Then

$$\langle Tz, z\rangle = \langle Tx, x\rangle + \langle Tx, y\rangle + \langle Ty, x\rangle + \langle Ty, y\rangle.$$

Also,

$$\langle z, Tz\rangle = \langle x, Tx\rangle + \langle x, Ty\rangle + \langle y, Tx\rangle + \langle y, Ty\rangle.$$

Since $\langle Tz, z\rangle = \langle z, Tz\rangle$, and also

$$\langle Tx, x\rangle = \langle x, Tx\rangle, \qquad \langle Ty, y\rangle = \langle y, Ty\rangle,$$

we get

$$\langle Tx, y\rangle + \langle Ty, x\rangle = \langle x, Ty\rangle + \langle y, Tx\rangle. \tag{1}$$

Now take $z = x + iy$. Then

$$\langle Tz, z\rangle = \langle Tx, x\rangle - i\langle Tx, y\rangle + i\langle Ty, x\rangle + \langle Ty, y\rangle.$$

Also,

$$\langle z, Tz\rangle = \langle x, Tx\rangle - i\langle x, Ty\rangle + i\langle y, Tx\rangle + \langle y, Ty\rangle.$$

Again using $\langle Tz, z\rangle = \langle z, Tz\rangle$ and canceling the diagonal terms, we get

$$-\langle Tx, y\rangle + \langle Ty, x\rangle = -\langle x, Ty\rangle + \langle y, Tx\rangle. \tag{2}$$

Adding (1) and (2), we obtain

$$2\langle Ty, x\rangle = 2\langle y, Tx\rangle.$$

Hence

$$\langle Tx, y\rangle = \langle x, Ty\rangle \qquad \text{for all } x, y \in \mathcal{H}.$$

Therefore, ($3 \implies 2$), and the three statements are equivalent. □

Corollary 6. *Let T be a bounded linear operator on a Hilbert space $\mathcal{H}$. Then T is* ***self-adjoint*** *if and only if $\langle x, Tx\rangle$ is* ***real*** *for all $x \in \mathcal{H}$.*

The following lemma will be used for The Spectum of an Operator section.

Lemma 2. *If T is self-adjoint, T is closed.*

0.5 Unitary operators

Definition 4. *An operator $T \in B(H)$ called unitary, if $TT^* = T^*T = I$.*

Example 17. *It can be see that the identity operator, I, is unitary, since $I^*I = II = I = II = II^*$.*

Example 18.

1. *$T_k : L^2[0,1] \to L^2[0,1]$, defined by $(T_k f)(t) = k(t)f(t)$, where $|k(t)| = 1$ for all $t \in [0,1]$.*
2. *The Fourier transform $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ is unitary. With the convention*

$$(\mathcal{F}f)(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x)e^{-ix\xi}\, dx,$$

we have $\mathcal{F}^* = \mathcal{F}^{-1}$.

Proposition 5. *Let $T \in B(H)$, then T is unitary if and only if T is surjective isometry.*

Proof. If T is unitary, then it is surjective, since it is invertible, and it is isometry, since

$$\|Tx\|^2 = \langle Tx, Tx\rangle = \langle x, T^*Tx\rangle = \langle x, x\rangle = \|x\|^2$$

Conversely, let T be a surjective isometry, we have

$$\begin{aligned}\langle x, T^*Tx\rangle = \langle Tx, Tx\rangle = \|Tx\| = \|x\| = \langle x, x\rangle &= \langle x, Ix\rangle \\ \implies \langle x, (T^*T - I)x\rangle &= 0\end{aligned}$$

Hence, $(T^*T - I)x = 0$ for all $x \in H$. This implies that $T^*T - I = 0 \implies T^*T = I$. Since T is surjective, for all y, there is x such that $Tx = y$. So,

$$TT^*y = TT^*(Tx) = T(T^*T)x = TIx = Tx = y$$

So, $TT^* = T^*T = I$. Therefore, T is unitary. Therefore, T is unitary. ☐

Theorem 11. *Let T, S be unitary operators, then*

1. *T^* is unitary and $\|T\| = \|T^*\| = 1$*
2. *TS is unitary*
3. *T^{-1} is unitary.*

Proof.

1. We know that $T^*T = I \implies T^{-1} = T^*$. So,

$$(T^*)^{-1} = (T^{-1})^* = (T^*)^* \implies (T^*)^*T^* = I$$

2. We have

$$(TS)^*TS = S^*T^*TS = S^*IS = S^*S = I$$

3. We have $(T^{-1})^*T^{-1} = (T^*)^*T^{-1} = TT^{-1} = I$

$\square$

Remark 1. *By Theorem 11, Example 17 and the associativity of function composition, we observe that the set of all unitary operators forms a group under composition. Moreover, if $T : \mathbb{C}^n \to \mathbb{C}^n$, this group is called the* ***unitary group****, denoted $U(n)$. There is a special subgroup of $\mathrm{U}(n)$, which is the* ***special unitary group****, denoted $\mathrm{SU}(n)$, which is the set of all unitary operators from $\mathbb{C}^n$ to $\mathbb{C}^n$ such that its determinant is 1. This forms a subgroup since if we let $A, B \in \mathrm{SU}(n)$, we have*

$$\det AB^{-1} = \frac{\det A}{\det B} = \frac{1}{1} = 1$$

showing that $AB^{-1} \in \mathrm{SU}(n)$.

0.6 The Spectrum of an Operator

The notation of an eigenvalue of a matrix has many essential applications in finite dimensional vector space; however, we wish we could extend this notation to the infinite-dimensional vector space.

Definition 5. *Let T be a linear operator, $\lambda \in \mathbb{C}$ is called an eigenvalue of T, if there exists $x \in H$ with $x \neq 0$ such that*

$$Tx = \lambda x$$

Definition 6. *Let T be a linear operator. The resolvent set of T, denoted by $\rho(T)$, is $\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is invertible}\}$. The spectrum of T, denoted by $\sigma(T)$ is $\sigma(T) = \mathbb{C} \backslash \rho(T)$, or equivalently, $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$*

Remark 2. *Notice that, if $T \in B(H)$, by Theorem 5, we can rewrite the resolvent set of T as*

$$\rho(T) = \{\lambda \in \mathbb{C} : \ker(T - \lambda I)^* = \{0\} \text{ and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \alpha > 0\}$$

Alternatively, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \|(T - \lambda I)^*x\| \geq \beta\|x\| \text{ and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \beta, \alpha > 0\}$$

by Corollary 4

Theorem 12. *If λ is an eigenvalue of T , then $\lambda \in \sigma(T)$*

Proof. Suppose that $\lambda \notin \sigma(T)$, then $T - \lambda I$ is invertible. So, we have $(T - \lambda I)u = 0 \implies u = (T - \lambda I)^{-1}(T - \lambda I)u = (T - \lambda I)^{-1}(0) = 0$. So, if there is $u \in \mathcal{H}$ such that $Tu = \lambda u$, then $u = 0$. So, λ is not an eignvalue of T. $\square$

Theorem 13. *Let H be a finite dimensional Hilbert space, and let $T : H \to H$ be linear. If $\lambda \in \sigma(T)$, then λ is an eigenvalue of T.*

Proof. Suppose that λ is not an eigenvalue of T, then there is nonzero $u \in H$ such that $Tu = \lambda u$. So, there is no nonzero $u \in H$ such that $(T - \lambda I)u = 0$. So, $\ker(T - \lambda I) = \{0\}$. By rank-nullity theorem, we have

$$\begin{aligned}\dim H &= \dim\ker(T - \lambda I) + \dim \operatorname{Im}(T - \lambda I)\\ &= 0 + \dim \operatorname{Im}(T - \lambda I)\\ &= \dim \operatorname{Im}(T - \lambda I)\end{aligned}$$

So, $\operatorname{Im}(T - \lambda I) = H$. This means that $(T - \lambda I)$ is surjective, and therefore, invertible. So, $\lambda \in \rho(T)$, and therefore, $\lambda \notin \sigma(T)$. $\square$

Remark 3. *For finite dimesonial Hilbert space, the spectrum of T consist only the eigenvalues of T. However, in infinite dimesional Hilbert space, we may have no eigenvalues. For instance, consider the shift operator defined in Example 8. Here, it has no eigenvalues, since if we consider $(T - \lambda I)x = 0$, we see that*

$$(-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \cdots) = (0, 0, 0, \cdots) \implies x = (0, 0, 0, \cdots)$$

So, T does not have an eigenvalue. However, in Example 9, we showed that T is not invertible. So, $T - 0I$ is not invertible and $0 \in \sigma(T)$. So, the spectrum of an operator does not only consist of eigenvalues.

Proposition 6. *Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \to \mathcal{H}$ be linear operator. The following properties hold:*

1. $\sigma(I) = \{1\}$
2. $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
3. *If T is invertible, then* $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$
4. *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*
5. *If $U : \mathcal{H} \to \mathcal{H}$ is invertible, then $\sigma(UTU^{-1}) = \sigma(T)$*

Proof.

1. Suppose that $\lambda \in \sigma(I)$, then $I - \lambda I = (1 - \lambda)I$ is not invertible. $(1 - \lambda)I$ is not invertible if and only if $1 - \lambda = 0$, since $1 - \lambda \in \mathbb{F}$. So, this is true if and only if $\lambda = 1$.

2. Suppose that $\mu \in \sigma(T^*)$, then $T^* - \mu I$ is not invertible. Observe that $T^* - \mu I$ is invertible if and only if $(T^* - \mu I)^* = T - \bar{\mu} I$ is invertible, since if $(T^* - \mu I)^*$ is invertible, then $((T^* - \mu I)^*)^* = T^* - \mu I$ is invertible. So, $T^* - \mu I$ is not invertible iff $(T^* - \mu I)^* = T - \bar{\mu} I$ is not invertible. So, $\bar{\mu} \in \sigma(T)$, i.e. $\mu \in \{\bar{\lambda} : \lambda \in \sigma(T)\}$.

3. If T is invertible, i.e. $T - 0I$ is invertible, then $0 \notin \sigma(T)$. Similarly, $0 \notin \sigma(T^{-1})$. Suppose that $\mu \in \sigma(T^{-1})$, then $T^{-1} - \mu I$ is not invertible. Observe that $T^{-1} - \mu I$ is invertible if and only if $T(T^{-1} - \mu I) = I - \mu T$ is invertible, since if $T(T^{-1} - \mu I)$ has an inverse, K, then KT would be an inverse of $T^{-1} - \mu I$. So, $T^{-1} - \mu I$ is not invertible iff $T(T^{-1} - \mu I) = I - \mu T$ is not invertible. Also, $I - \mu T$ is not invertible iff $T - \mu^{-1} I$, where μ^{-1} exists, since $\mu \neq 0$. So, $\mu^{-1} \in \sigma(T)$, i.e. $\mu \in \{\lambda^{-1} : \lambda \in \sigma(T)\}$.

4. If T is self-adjoint, then if $a \in \mathbb{R}$, then $T - aI$ is self-adjoint. This is since $(T - aI)^* = T^* - \bar{a}I = T - aI$. So, $\langle (T - aI)x, x\rangle \in \mathbb{R}$. Observe that, if $\lambda = a + ib$, then

$$\begin{aligned}\|(T-\lambda I)x\|\|x\| \geq |\langle (T-\lambda I)x, x\rangle| &= |\langle (T-aI)x - ibI(x), x\rangle| \\ &= |\langle (T-aI)x, x\rangle - ib\langle x, x\rangle| \\ &= \sqrt{\langle (T-aI)x, x\rangle^2 + b^2\langle x, x\rangle^2} \\ &\geq |b|\langle x, x\rangle \\ &= |b|\|x\|^2\end{aligned}$$

So,

$$\|(T-\lambda I)x\| \geq |b|\|x\|$$

Similarly,

$$\|(T-\lambda I)^* x\| \geq |b|\|x\|$$

So, $T - \lambda I$ is injective, since $(T-\lambda I)x = 0 \implies x = 0$, and $\ker(T-\lambda I)^* = \{0\}$. So, $\operatorname{Im}(T-\lambda I)$ is dense in $\mathcal{H}$. This means that for all $y \in \mathcal{H}$, there is a sequence $\{(T-\lambda I)x_n\} \in \operatorname{Im}(T-\lambda I)$ such that $(T-\lambda I)x_n \to y$. So, $\{(T-\lambda I)x_n\}$ is a Cauchy sequence and $\{x_n\}$ is Cauchy, since

$$\|x_n - x_m\| \leq \frac{1}{|b|}\|(T-\lambda I)x_n - (T-\lambda I)x_m\|$$

Since $\mathcal{H}$ is a Hilbert space, $\{x_n\}$ converges to some $x \in \mathcal{H}$. So, $(T-\lambda I)x_n + \lambda x_n \to y + \lambda x$. Since T is self adjoint, the T is closed, by **??**. By closedness of T, $y = (T-\lambda I)x + \lambda x = y + \lambda x$, i.e. $(T\lambda I)x \to y$. So, $T - \lambda I$ is surjective and $\lambda \in \rho(T)$ if $b \neq 0$. So, $\sigma(T) \subseteq \mathbb{R}$

5. Suppose that $\lambda \in \sigma(UTU^{-1})$, then $UTU^{-1} - \lambda I$ is not invertible. Observe that $UTU^{-1} - \lambda I = U(T-\lambda)U^{-1}$ is not invertible iff $T - \lambda I$ is not invertible. So, $\lambda \in \sigma(T)$.

□

Theorem 14. *Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$. If $|\lambda| > \|T\|$ then $\lambda \notin \sigma(T)$.*

Proof. Let $u \in \ker(T-\lambda I)^*$, then $(T-\lambda I)^* u = (T^* - \bar{\lambda} I)u = 0$. So, $T^* u - \bar{\lambda} u = 0 \implies T^* u = \bar{\lambda} u$. So,

$$|\lambda|\|u\| = |\bar{\lambda}|\|u\| = \|\bar{\lambda} u\| = \|T^* u\| \leq \|T^*\|\|u\| = \|T\|\|u\|$$

$$\implies (|\lambda| - \|T\|)\|u\| \leq 0$$

Since $|\lambda| > \|T\|$, then $\|u\| \leq 0 \implies \|u\| = 0 \implies u = 0$. Hence, $\ker(T-\lambda I)^* = \{0\}$. We also have that

$$\begin{aligned}\|(T-\lambda I)x\| = \|Tx - \lambda x\| \geq |\|Tx\| - |\lambda|\|x\|| &= \left|\frac{\|Tx\|}{\|x\|} - |\lambda|\right|\|x\| \\ &= \left(|\lambda| - \frac{\|Tx\|}{\|x\|}\right)\|x\| \\ &\geq (|\lambda| - \|T\|)\|x\| \\ &= \alpha\|x\|\end{aligned}$$

where $\alpha := |\lambda| - \|T\|$ is positive, since $|\lambda| > \|T\|$. So, $\lambda \in \rho(T)$, by first equality in Remark 2. Hence, $\lambda \notin \sigma(T)$. □

Corollary 7. *If $T \in B(H)$, then $\sigma(T)$ is bounded.*

Proof. If $\lambda \in \sigma(T)$, then, by the contrapositive of Theorem 14, $|\lambda| \leq \|T\|$. So, $\lambda \in \overline{B(0, \|T\|)}$. Hence, $\sigma(T) \subseteq \overline{B(0, \|T\|)}$, i.e. $\sigma(T)$ is bounded. $\square$

Corollary 8. *If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then*

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Proof. If $|\lambda| > 1 = \|U\|$, then $\lambda \notin \sigma(U)$, by Theorem 14. Moreover, if $|\lambda| < 1$, then

$$U - \lambda I = -\lambda U(U^{-1} - \frac{1}{\lambda} I)$$

is invertible, since $-\lambda U$ is invertible and, by Theorem 14, $\left|\frac{1}{\lambda}\right| > 1 = \|U^{-1}\| \implies \frac{1}{\lambda} \notin \sigma(U^{-1})$. So, $\lambda \notin \sigma(U)$. Therefore, $|\lambda| = 1$ if $\lambda \in \sigma(U)$. $\square$

Theorem 15. *If $T \in B(H)$, then $\sigma(T) \subseteq \mathbb{C}$ is a closed set.*

Proof. We will show that $\rho(T) = \mathbb{C} \backslash \sigma(T)$ is open. Let $\lambda \in \rho(T)$. This means that, by second equality in Remark 2, there is $\alpha > 0$ such that

$$\|(T - \lambda I)u\| \geq \alpha \|u\| \implies \frac{\|(T - \lambda I)u\|}{\|u\|} \geq \alpha$$

Since the quantity $\frac{\|(T-\lambda I)u\|}{\|u\|}$ has a lower bound, its infimum exists. Similarly, by second equality in Remark 2, the infimum of the quantity $\frac{\|(T-\lambda I)^* u\|}{\|u\|}$ exists. Now, let

$$\gamma := \inf_{u \in H \backslash \{0\}} \frac{\|(T - \lambda I)u\|}{\|u\|}$$
$$\gamma' := \inf_{u \in G \backslash \{0\}} \frac{\|(T - \lambda I)^* u\|}{\|u\|}$$

and let $\delta := \frac{1}{2} \min(\gamma, \gamma')$. We want to show that

$$B(\lambda, \delta) \subseteq \rho(T)$$

Let $\tilde{\lambda} \in B(\lambda, \delta)$, then $|\tilde{\lambda} - \lambda| < \delta$. First, we observe that

$$\begin{aligned}
\|(T - \tilde{\lambda} I)u\| = \|(T - \lambda I)u - (\tilde{\lambda} - \lambda)I(u)\| &\geq \big|\|(T - \lambda I)u\| - |\tilde{\lambda} - \lambda|\|u\|\big| \\
&= \Big|\frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda|\Big|\|u\| \\
&= \Big(\frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda|\Big)\|u\| \\
&> (\gamma - \frac{1}{2}\min(\gamma, \gamma'))\|u\| \\
&\geq (\gamma - \frac{\gamma}{2})\|u\| \\
&= \frac{\gamma}{2}\|u\|
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
\|(T-\tilde{\lambda}I)^*u\| = \|(T-\lambda I)^*u - \overline{(\tilde{\lambda}-\lambda)}I(u)\| &\geq |\|(T-\lambda I)^*u\| - |\overline{\tilde{\lambda}-\lambda}|\|u\|| \\
&= |\|(T-\lambda I)^*u\| - |\tilde{\lambda}-\lambda|\|u\|| \\
&= \Big|\frac{\|(T-\lambda I)^*u\|}{\|u\|} - |\tilde{\lambda}-\lambda|\Big|\|u\| \\
&= \Big(\frac{\|(T-\lambda I)^*u\|}{\|u\|} - |\tilde{\lambda}-\lambda|\Big)\|u\| \\
&> (\gamma' - \frac{1}{2}\min(\gamma,\gamma'))\|u\| \\
&\geq (\gamma' - \frac{\gamma'}{2})\|u\| \\
&= \frac{\gamma'}{2}\|u\|
\end{aligned}$$

So, by the second equality in Remark 5, $\tilde{\lambda} \in \rho(T)$. Therefore, $B(\lambda,\delta) \subseteq \rho(T)$. □

Corollary 9. *If $T \in B(H)$, then $\sigma(T)$ is compact in $\mathbb{C}$*

Proof. By Theorem 15 and Corollary 7, $\sigma(T)$ is closed and bounded. By Heine-Borel Theorem, $\sigma(T) \subseteq \mathbb{C}$ is compact. □

Example 19. *Let $T : \ell^2 \to \ell^2$ be the unilateral shift operator defined in Example 8. We would like to find the spectrum of T, $\sigma(T)$. First, we claim that if $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then λ is an eigen value of T^*. Consider the sequence*

$$x = (1, \lambda, \lambda^2, \cdots)$$

Here, $x \in \ell^2$, since

$$\sum_{n=0}^{\infty} |x_n|^2 = \sum_{n=0}^{\infty} (|\lambda|^n)^2 = \sum_{n=0}^{\infty} (|\lambda|^2)^n = \frac{1}{1-|\lambda|^2} < \infty$$

where this series converge, since $|\lambda| < 1$. Now, observe that

$$\begin{aligned}
Tx &= T(1,\lambda,\lambda^2,\cdots) \\
&= (\lambda,\lambda^2,\lambda^3,\cdots) = \lambda(1,\lambda,\lambda^2,\cdots) = \lambda x
\end{aligned}$$

So, $x \in \ell^2$ is an eigenvalue of T. This means that $\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T^*)$. We have $\{\lambda \in \mathbb{C} : |\lambda| < 1\} = \{\bar{\lambda} : |\lambda| < 1\} \subseteq \{\bar{\lambda} : \lambda \in \sigma(T^*)\} = \sigma(T)$. So, by theorem 14 and that T is an isometry, we have

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

By theorem 15, $\sigma(T)$ is closed. Therefore,

$$\sigma(T) = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

Example 20. *Let $a \in \ell^\infty$, define $D_a : \ell^2 \to \ell^2$ by*

$$D_a(x_1, x_2, \cdots) = (a_1x_1, a_2x_2, \cdots)$$

Fisrt, note that D_a is a bounded linear operator, since

$$\begin{aligned}\|D_a x\| = \Big(\sum_{n=1}^{\infty} |a_n x_n|^2\Big)^{\frac{1}{2}} = \Big(\sum_{n=1}^{\infty} |a_n|^2 |x_n|^2\Big)^{\frac{1}{2}} &\le \Big(\sup |a_n|^2 \sum_{n=1}^{\infty} |x_n|^2\Big)^{\frac{1}{2}} \\ &= \sup |a_n| \Big(\sum_{n=1}^{\infty} |x_n|^2\Big)^{\frac{1}{2}} \\ &= \sup |a_n| \|x\|\end{aligned}$$

Now, we would like to find $\sigma(D_a)$. First, we will begin by finding its eigenvalues in order to find a lower bound for the set. Suppose that λ is an eigenvalue of D_a, then we have that $D_a - \lambda I$ is

$$(D_a - \lambda I)(x_1, x_2, \cdots) = ((a_1 - \lambda)x_1, (a_2 - \lambda)x_2, \cdots) = (0, 0, \cdots)$$

if either $x_n = 0$ for all n or $a_k - \lambda = 0$ for some k. For the former case, this yields $x = 0$. For the later case, $\lambda = a_k$ for some k. So, if λ is an eigenvalue of D_a, then $\lambda \in \{a_n | n \in \mathbb{N}\}$. Indeed, if $e_k = (0, \cdots, 0, 1, 0, \cdots)$, with 1 on the kth term of the sequence, then we have

$$\begin{aligned}D_a e_k &= D_a(0, \cdots, 0, 1, 0, \cdots) \\ &= (0, \cdots, 0, a_k, 0, \cdots) = a_k(0, \cdots, 0, 1, 0, \cdots) = a_k e_k\end{aligned}$$

So, the set of eigenvalues of D_a is $\{a_n | n \in \mathbb{N}\}$. Hence, $\{a_n | n \in \mathbb{N}\}$. Since $\sigma(D_a)$ is a closed set, we have $\overline{\{a_n | n \in \mathbb{N}\}} \subseteq \sigma(D_a)$.
We will now show that $\sigma(D_a) \subseteq \overline{\{a_n | n \in \mathbb{N}\}}$. Observe that

$$\begin{aligned}\|(D_a - \lambda I)x\| = \|(a_1 - \lambda)x_1, (a_2 - \lambda)x_2, \cdots\| &= \Big(\sum_{n=1}^{\infty} |(a_n - \lambda)x_n|^2\Big)^{\frac{1}{2}} \\ &= \Big(\sum_{n=1}^{\infty} |a_n - \lambda|^2 |x_n|^2\Big)^{\frac{1}{2}} \\ &\ge \Big(\inf |a_n - \lambda|^2 \sum_{n=1}^{\infty} |x_n|^2\Big)^{\frac{1}{2}} \\ &= \inf |a_n - \lambda| \Big(\sum_{n=1}^{\infty} |x_n|^2\Big)^{\frac{1}{2}} \\ &= \inf |a_n - \lambda| \|x\|\end{aligned}$$

where $|a_n - \lambda|$ is bounded since a_n is bounded. Similarly, we have

$$\|(D_a - \lambda I)^* x\| \ge \inf |a_n - \lambda| \|x\|$$

We see that if $D_a - \lambda I$ is not invertible, then $\inf |a_n - \lambda| = 0$. If $\inf |a_n - \lambda|$, for all $\epsilon > 0$, there is $k \in \mathbb{N}$ such that $|a_k - \lambda| < \inf |a_k - \lambda| + \epsilon = \epsilon$. So, there is $a_k \in \overline{\{a_n | n \in \mathbb{N}\}} \cap B(\lambda, \epsilon)$, i.e. $\{a_n | n \in \mathbb{N}\} \cap B(\lambda, \epsilon) \neq$, for all $\epsilon > 0$. So, $\lambda \in \overline{\{a_n | n \in \mathbb{N}\}}$. Therefore, $\sigma(T) = \overline{\{a_n | n \in \mathbb{N}\}}$.